Airline Data Analysis Report (2021-2023)

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1. Dataset Description:

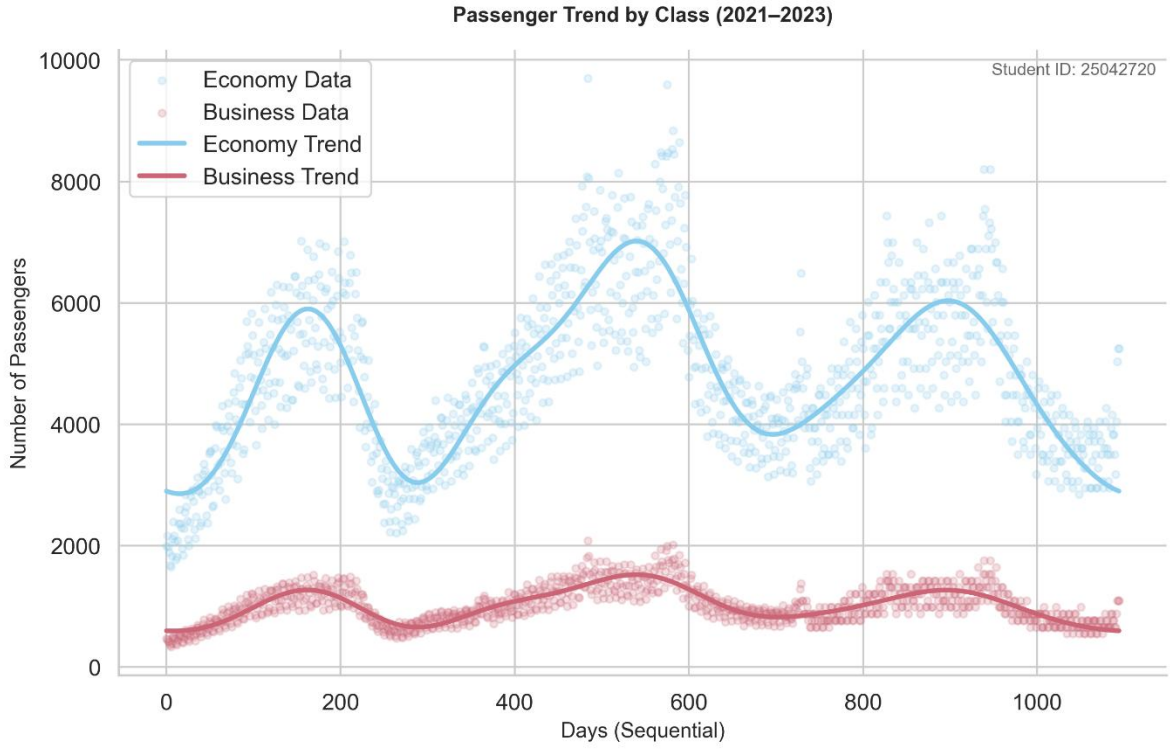
This analysis is depended on two dataset which is about airline data of the year 2021 to 2023. In which first dataset contains data of 2021-2022 daily observations. Which has date, passenger counts and price of tickets for Domestic, Europe and Intercontinental and which further parts into two sub category Business and Economy class. Also, it has another column name AS revenue which indicates the daily income of extra services. Where the second dataset is sample of a dataset which provided the detailed transaction of passengers in which it has a date, ticket price, extra pay for service, class like Business(J) or Economy(Y) and Region like Domestic, Europe and Intercontinental.

The dataset combines the time-series data for analysis of daily and monthly trends, categorical variables like class, region and the numerical value for calculate the revenue and passenger traffic trends.

2. Visual Analysis

Figure 1: Passenger Trend by Class (2021-2023):

Passenger numbers in daily basis into subcategory in Economy and business class with smoothed trends.



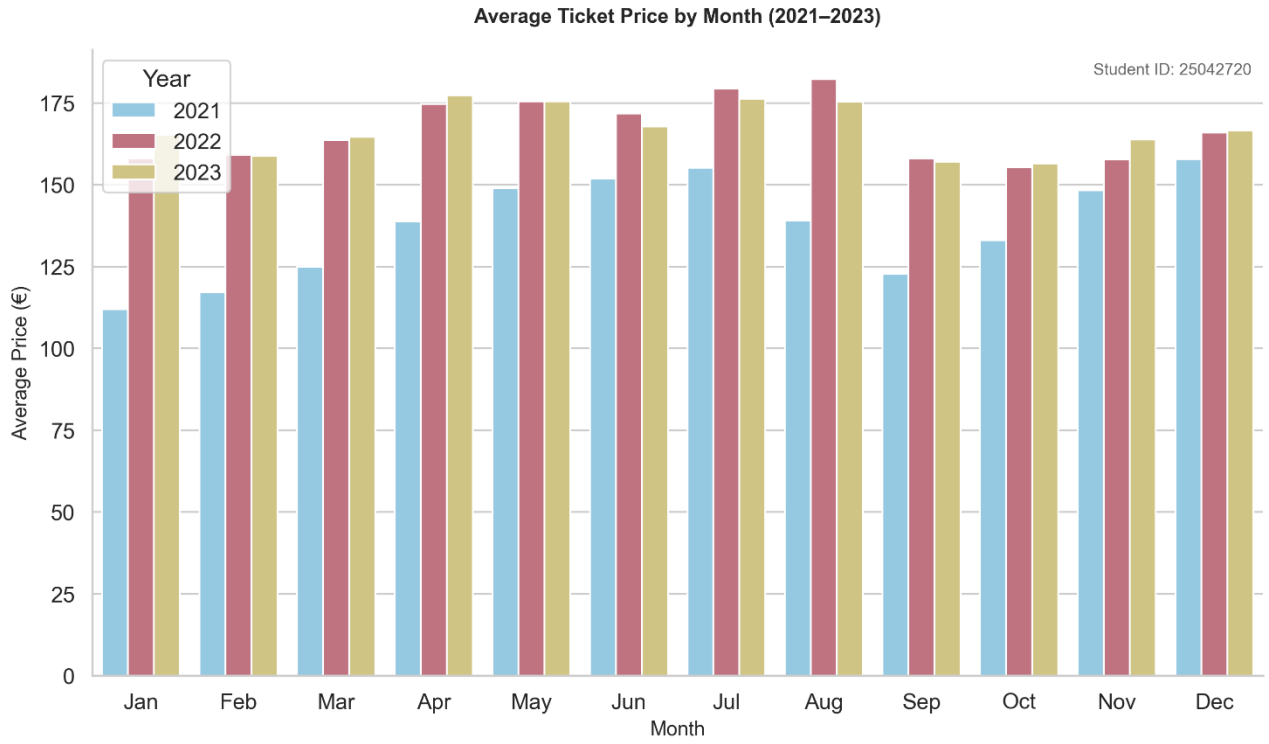
Truncated Equation of Fourier Series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left(a_n \cos\left(\frac{2\pi n x}{L}\right) + b_n \sin\left(\frac{2\pi n x}{L}\right) \right)$$

This equation used to model periodic passenger trends and for the smooth line specifically used first 8 terms of Fourier series so that means N=8. It gives smooth seasonal patterns trend line.

Figure 2: Average Ticket Price by Month (2021-2023):

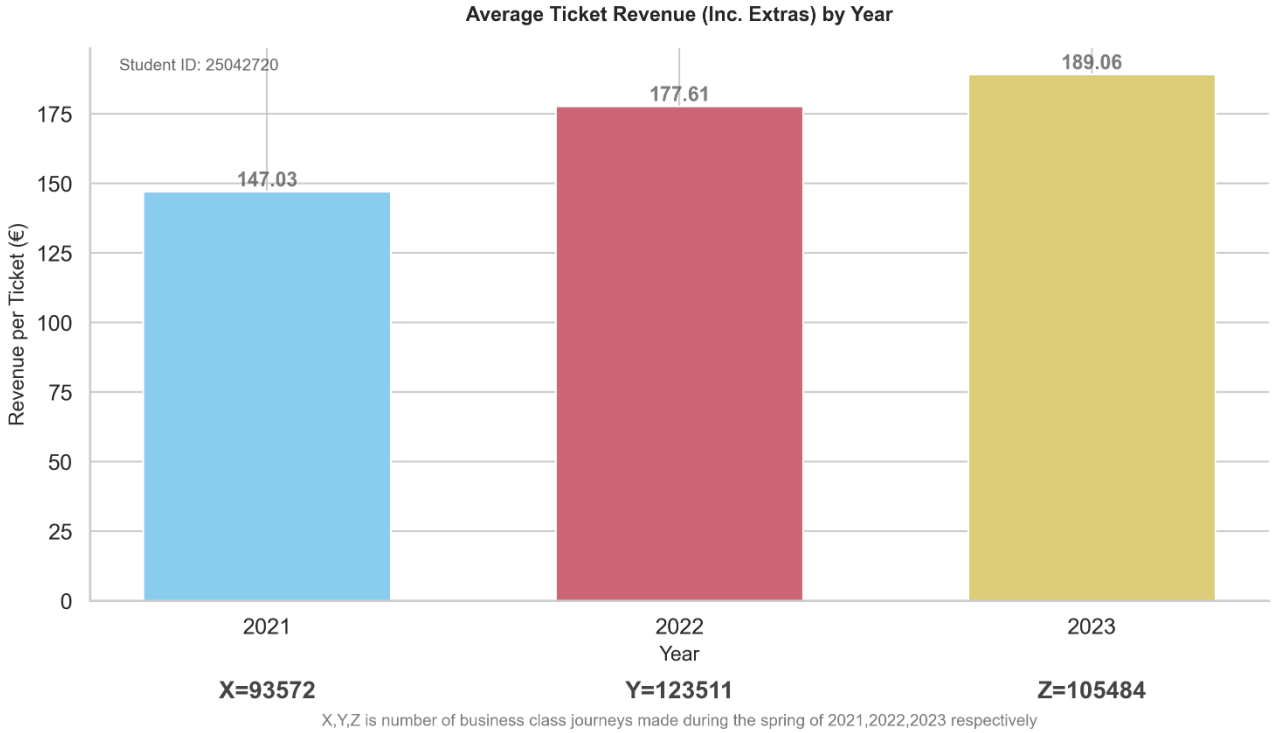
Monthly Comparison of average ticket price across three years



Monthly Average Price of Tickets: $P = \frac{1}{n}\sum P_i$

This formula used to calculate average ticket prices per month. Where $\mathbf{P_i}$ is daily average price of tickets and **n** is number of days in particular month.

Figure 3: Average Ticket Revenue Including Extra Services by Year:



Annual comparison of ticket prices including extra revenue per ticket across three years.

Average Revenue Model:

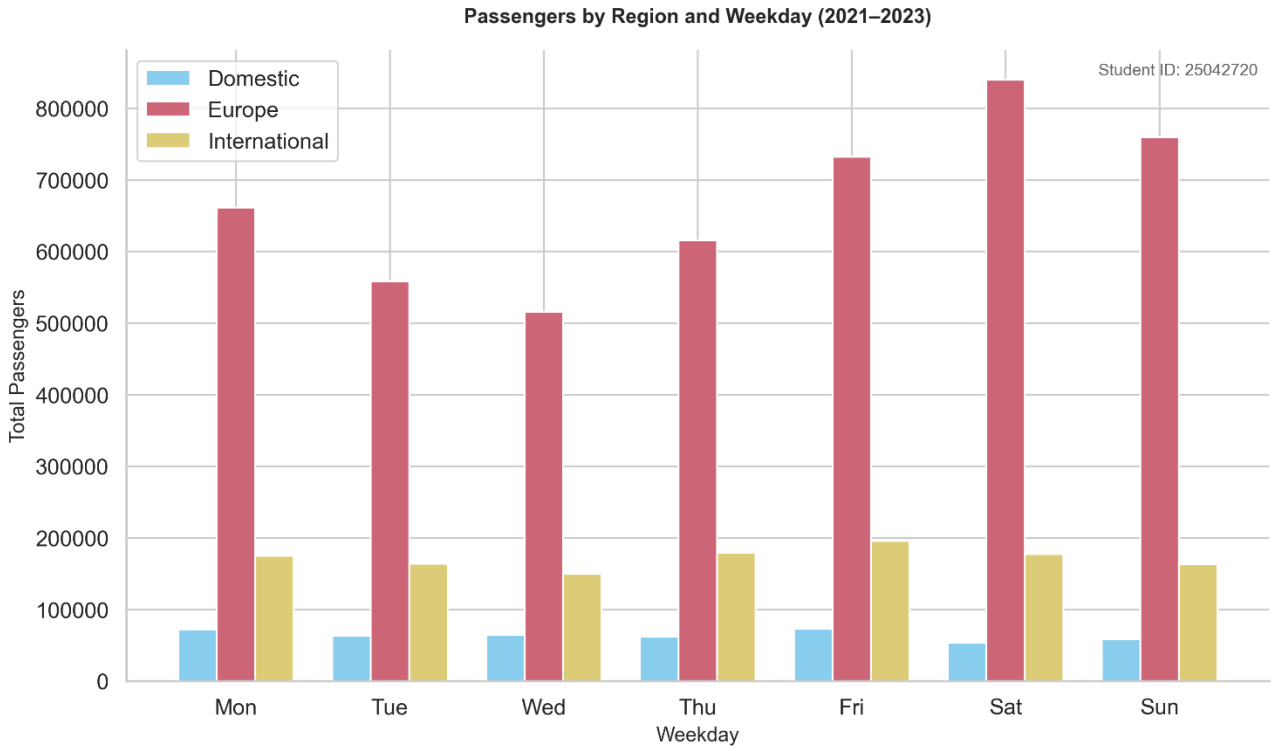
$$R = P \times N + E$$

Where P is ticket price and N is number of passenger and E is extra revenue. Then again used average formula to calculate price by years.

Were, X, Y, Z calculated by sum of the total passenger in business class during spring months.

Figure 4: Passenger by Region and Weekday (2021-2023):

Passenger distribution by weekday and further in sub-category by region.



3. Discussion & Interpretation:

The visual analysis of data depicts that there are significant trends in passenger travel, price correction and revenue performance between 2021 to 2023.

Passenger travel trends are **seasonality**; it increases strongly during mid-years that reflects travel behaviour of passenger during holidays and summer break where in weekly basis passenger travel peak on weekends which also support their travel pattern in their leisure time. Most of the passenger prefer to travel in Economy class and which dominates the trend of seasonality while Business class passenger remain stable during whole year. It happening may be because of corporate travel and its price while Economy class may be mostly chosen by family and normal travellers.

Ticket price sharply increased in 2022 in respect of 2021, while the minimum average price of tickets to remain airline profitable increase only 1 Euro and the average ticket price increase by ~30 Euro which shows that the moves take by airline to increase their profit share or may be recover some past loss like the past-pandemic time. Despite of the price hike of ~19% the passenger demands remain strong. Where in 2023 price remain stabilize, indicates that some strong competition or

market stability. Each year, airline increases their price in the mid-year while there is high demand of the passengers which is the profitable move for airlines to increase their profit share and also maintain the capacity of passenger handling.

Price analysis of the data shows that in 2022 price increased and also passenger increased from the past year, in 2023 cost increased and passenger still increased that shows Inelastic demand of travel. It shows customer continue to travel when price rise may be because of essential travelling, limited airline competition. Where the sharp cost price increase in 2023 from 79 Euro to 121 Euro suggests the possibility of rising fuel price, inflation and labour and operational expenses.

In figure 3 value X, Y, Z shows the total number of passenger travel in business class during spring in 2021,2022,2023 respectively. It shows that in spring 2022 passenger number is 123,511, which increase in the respect of year spring 2021 where the number of passengers is only 93,572, which shows the demand of passenger increased by 25% while in 2023 the number of business class passenger decrease with 14% and the passenger number during that spring is 105,484.

In regional analysis, European routes dominate the largest number of the passenger rather than Domestic routes and Intercontinental routes. Which make European routes primary routes for revenue generate. And then Intercontinental routes have more passenger number than Domestic Routes which shows that it happens because of people may prefer to travel by other transportation services on domestic level.

The relationship between the price, passenger numbers and revenue from the analysis shows that it is driven by the combination of pricing and passenger demand. Airline successfully increase their price by 30% in 2022 without any reduced demand of passenger indicating their strong and successful move. While talk about the distance it highly depends on the region. Talk about Domestic and Europe routes almost all price lies lesser than 250 and 500 Euro Respectively but, in Intercontinental routes price depends on the distance when the distance increase price also increased linearly and it increased in two different trend line one for Economy class and other for Business. When distance increased, price growth of Business class in Intercontinental journey increase sharply rather than Economy class.

In year 2021, the total passenger travel around 19 million with average ticket price of 147.03 Euro where the minimum ticket price to remain airline profitable is 78 Euro and that year airline generated revenue ~2,794.60 million Euro and they earn profit of ~1,312.06 million Euro which shows their profit margin is around 46.95% in that year.

In year 2022, total passenger is around 24 million with average ticket price of 177.61 Euro and the minimum ticket price is 79 Euro and the generated revenue is ~4,274.00 million Euro and profit is ~2372.96 million Euro so the profit margin is around 55.52% in that year.

In year 2023, with 25 million of travel passengers and with average ticket price of 189.06 Euro where minimum ticket price is 121 Euro airline generated revenue of ~4,804.14 million Euro and they earn profit of ~1728.40 million Euro with around 36.00% of profit margin.

From the profit perspective, year 2021 represents that it has in some recovery phase where the passenger demands have lower and so that price also lower and the revenue is least rather than other two year. Year 2022 appears to be the most profitable year against the other two year, it has attended the highest number of passengers in that year and the price also be higher, the revenue of that year is lower than year 2023 but the profit share is higher because of the lower cost than the year 2023. That’s why despite of the higher revenue airline earned lesser profit than 2022.

Final Conclusion:

The airline has successfully transitioned from recovery in 2021 to its peak performance in 2022 and then high-cost growth phase 2023. While demand remains high, increasing cost price it has in high risk for future profitability. So, to stay profitable it will need to find a way of cost control strategy and pricing strategy.